Colligative properties of electrolyte

21. Which of the following has lowest freezing point

(a) $0.1M$ aqueous solution of glucose

(b) $0.1M$ aqueous solution of $NaCl$

(c) $0.1M$ aqueous solution of $ZnSO_4$

(d) $0.1M$ aqueous solution of urea

22. The freezing points of equimolar solutions of glucose, KNO_3 and $AlCl_3$ are in the order of

(a) $AlCl_3 < KNO_3 < \text{Glucose}$

(b) $\text{Glucose} < KNO_3 < AlCl_3$

(c) $\text{Glucose} < AlCl_3 < KNO_3$

(d) $AlCl_3 < \text{Glucose} < KNO_3$

23. Which of the following will have the highest F.P. at one atmosphere

(a) $0.1M NaCl$ solution

(b) $0.1M$sugar solution

(c) $0.1M BaCl_2$ solution

(d) $0.1M FeCl_3$ solution

24. Which of the following will produce the maximum depression in freezing point of its aqueous solution

(a) $0.1M$ glucose

(b) $0.1M$sodium chloride

(c) $0.1M$barium chloride

(d) $0.1M$magnesium sulphate

25. Which of the following has the lowest freezing point

(a) 0.1 m sucrose (b) 0.1 m urea

(c) 0.1 m ethanol (d) 0.1 m glucose

26. Which of the following has minimum freezing point

(a) $0.1M$ $K_2Cr_2O_7$

(b) 0.1 M NH_4Cl

(c) 0.1 M $BaSO_4$

(d) 0.1 M $Al_2(SO_4)_3$

27. Which of the following $0.10m$ aqueous solution will have the lowest freezing point

(a) $Al_2(SO_4)_3$ (b) $C_5H_{10}O_5$

(c) KI (d) $C_{12}H_{22}O_{11}$

28. For 0.1 M solution, the colligative property will follow the order

(a) $NaCl > Na_2SO_4 > Na_3PO_4$

(b) $NaCl < Na_2SO_4 < Na_3PO_4$

(c) $NaCl > Na_2SO_4 \approx Na_3PO_4$

(d) $NaCl < Na_2SO_4 = Na_3PO_4$

29. Which of the following will have the lowest vapour pressure

(a) $0.1M KCl$ solution

(b) $0.1M$ urea solution

(c) $0.1M Na_2SO_4$ solution

(d) $0.1M K_4Fe(CN)_6$ solution